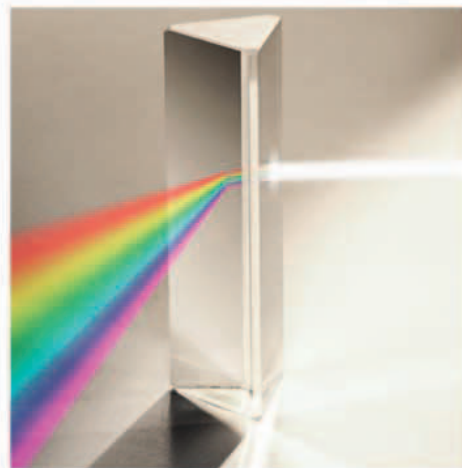
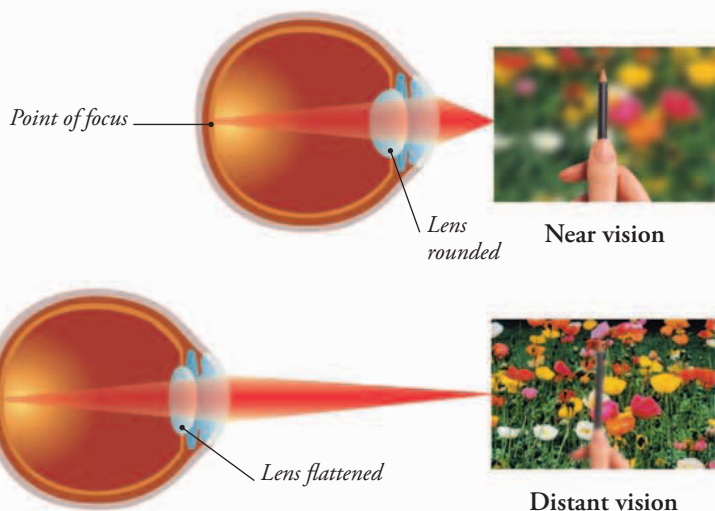


IN FOCUS

Eyes work much like cameras—they have to focus light with a lens to create a sharp image. Camera lenses do this by moving forward and back, but the lens in a human eye focuses by automatically changing its shape. When you want to focus on something near your eye, special muscles squeeze the lens into a more rounded shape. If you want to look at something in the distance, the muscles relax so the lens flattens and focuses farther away.



SEEING IN COLOR

The cone cells in the retina of each eye allow you to see in color. Different cells respond to various levels of red, blue, and green light. They send signals to your brain, which combines the data and lets you distinguish thousands of different colors. Some people are not able to see the difference between colors such as red and green. This is known as color blindness.



Image formed by right eye



NEAR- AND FARSIGHTED

Many people are nearsighted—they cannot focus on things in the distance. Older people often develop the opposite problem, and cannot focus close up enough to read. Both problems can be corrected by wearing glasses or contact lenses, which are placed directly on the eyes.

FILLING IN THE GAPS

Sometimes you recognize something you are looking at, even though you don't get a clear view of it. This is because your brain fills in the gaps between the visual clues. There is a spotted dog in this picture, but it is very difficult to see because so much visual information is missing. Can you see it? If so, it's because your brain is using memory to complete the picture.



WOW!

The farthest object that the human eye can see without a telescope is the Andromeda Galaxy, which is an amazing 2.25 million light-years away.